

Capacitance of a Tilted Circular Parallel-Plate Capacitor with Application to Precision Tilt Measurement

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1 Problem Statement

Precision tilt sensors based on capacitive measurement exploit the sensitivity of capacitance to small changes in the air gap between two conducting plates. In a practical instrument, a pendulum of length L suspends one plate so that its centre-to-centre distance to a fixed reference plate varies with the tilt angle α . When the instrument is tilted, the two plates are no longer parallel: they subtend the same angle α .

The goal of this note is to derive a closed-form expression for the capacitance as a function of the tilt angle, compare it with two progressively cruder approximations, and invert each expression to obtain α as a function of the measured capacitance.

2 Variable Definitions and Conventions

Symbol	Unit	Description
α	rad	Tilt angle; $\alpha = 0$ corresponds to parallel plates
L	m	Pendulum length (constant)
r	m	Radius of the circular capacitor plates (constant)
g	m	Neutral air gap at $\alpha = 0$ (constant, $g > 0$)
d_c	m	Centre-to-centre plate distance at angle α
ε_0	F/m	Permittivity of free space, $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m
C	F	Total capacitance
x	m	Signed distance from the plate centre along the tilt axis

By convention, $\alpha > 0$ means the pendulum swings in the positive x -direction, closing the gap on that side. Negative values of α are equally valid and correspond to the opposite direction.

3 Geometry and Physical Limits

3.1 Centre distance as a function of tilt

The pendulum geometry gives the centre distance

$$d_c(\alpha) = g - L \sin \alpha. \quad (1)$$

At $\alpha = 0$ the plates are parallel with uniform gap g . For $\alpha \neq 0$ the plates are tilted by the same angle α , so the local gap at a signed distance x from the plate centre (measured along the tilt axis) is

$$d(x, \alpha) = d_c(\alpha) + x \sin \alpha. \quad (2)$$

3.2 Physical validity limit

The plates must not touch anywhere. The smallest gap occurs at $x = -r$ (the trailing edge when $\alpha > 0$), so the condition $d(-r, \alpha) > 0$ gives

$$\boxed{\alpha < \arcsin\left(\frac{g}{L + r}\right)}. \quad (3)$$

4 Exact Capacitance

4.1 From the parallel-plate formula to an integral

The standard parallel-plate capacitor formula,

$$C = \varepsilon_0 \frac{A}{d}, \quad (4)$$

assumes a uniform gap d over the entire plate area A . When the gap varies with position, equation (4) is applied locally to each infinitesimal area element dA , and the total capacitance is obtained by integrating the local contribution $dC = \varepsilon_0 dA/d(x, y)$:

$$C = \varepsilon_0 \iint_{\text{disk}} \frac{dA}{d(x, y)}. \quad (5)$$

A double integral is required because the integration domain is a two-dimensional disk. Since the gap (2) depends only on x and not on y , the inner integral over y can be evaluated immediately. For the circular plate of radius r the y -limits at a given x are $\pm\sqrt{r^2 - x^2}$, giving

$$C(\alpha) = 2\varepsilon_0 \int_{-r}^{+r} \frac{\sqrt{r^2 - x^2}}{d_c + x \sin \alpha} dx. \quad (6)$$

4.2 Evaluation of the integral

Equation (6) is of the standard form

$$I = \int_{-r}^{+r} \frac{\sqrt{r^2 - x^2}}{a + bx} dx = \frac{\pi}{b^2} \left(a - \sqrt{a^2 - b^2 r^2} \right), \quad a > br, \quad (7)$$

with $a = d_c$ and $b = \sin \alpha$. Substituting and simplifying gives

$$C(\alpha) = \frac{2\pi\varepsilon_0}{(\sin \alpha)^2} \left(d_c - \sqrt{d_c^2 - r^2 (\sin \alpha)^2} \right). \quad (8)$$

This form is indeterminate (0/0) at $\alpha = 0$. Rationalising by multiplying numerator and denominator by $d_c + \sqrt{d_c^2 - r^2 (\sin \alpha)^2}$ eliminates the singularity:

$$\boxed{C(\alpha) = \frac{2\pi\varepsilon_0 r^2}{d_c + \sqrt{d_c^2 - r^2 (\sin \alpha)^2}}}, \quad (9)$$

with $d_c = g - L \sin \alpha$ from equation (1). Equation (9) is valid for all α satisfying the physical limit (3).

4.3 Verification at $\alpha = 0$

At $\alpha = 0$, $d_c \rightarrow g$ and $\sin \alpha \rightarrow 0$. Expanding the square root for small $s = \sin \alpha$,

$$\sqrt{d_c^2 - r^2 s^2} \approx d_c - \frac{r^2 s^2}{2d_c}, \quad (10)$$

so the denominator of (9) approaches $2d_c \rightarrow 2g$, giving

$$C(0) = \frac{\pi r^2 \varepsilon_0}{g} = \varepsilon_0 \frac{A}{g}, \quad (11)$$

which is the standard parallel-plate result with $A = \pi r^2$. ✓

5 Approximate Formulae

5.1 Approximation neglecting the tilt angle

If the plate radius r is small compared with the ratio $d_c/\sin\alpha$, the term $r^2(\sin\alpha)^2$ under the square root in (9) is negligible. Physically, this corresponds to treating the plates as parallel (using only the centre gap) while still accounting for the change in d_c with α . Setting $r\sin\alpha \approx 0$ in the denominator of (9) reduces it to $2d_c$:

$$C_{\text{approx}}(\alpha) = \frac{\pi r^2 \varepsilon_0}{d_c} = \frac{\pi r^2 \varepsilon_0}{g - L \sin \alpha}. \quad (12)$$

5.2 Linear approximation

A further simplification is obtained by linearising around $\alpha = 0$. The sensitivity is found by differentiating (12) with respect to C at the neutral point. Both the exact and approximate formulae coincide at $\alpha = 0$, so their sensitivities are identical there. Implicit differentiation gives

$$\left. \frac{d\alpha}{dC} \right|_{\alpha=0} = \frac{g^2}{\pi r^2 \varepsilon_0 L}, \quad (13)$$

and the linear approximation around $\alpha = 0$, $C = C_0$ is

$$\alpha_{\text{lin}} = \frac{g^2}{\pi r^2 \varepsilon_0 L} (C - C_0), \quad (14)$$

with $C_0 = \pi r^2 \varepsilon_0 / g$ from equation (11).

6 Inversion: Solving for α

In practice, the capacitance C is the measured quantity and the tilt angle α is the desired output. We solve each formula for α .

6.1 Linear approximation

Equation (14) is already solved for α :

$$\alpha_{\text{lin}} = \frac{g^2}{\pi r^2 \varepsilon_0 L} (C - C_0). \quad (15)$$

6.2 Approximate formula

Solving (12) for $\sin\alpha$ and applying the inverse sine:

$$\alpha_{\text{approx}} = \arcsin \left[\frac{1}{L} \left(g - \frac{\pi r^2 \varepsilon_0}{C} \right) \right]. \quad (16)$$

6.3 Exact formula

Starting from (9) and writing $s = \sin \alpha$, $d_c = g - Ls$, and $k = 2\pi\epsilon_0 r^2/C$:

$$\begin{aligned} d_c + \sqrt{d_c^2 - r^2 s^2} &= k \\ \sqrt{d_c^2 - r^2 s^2} &= k - d_c \\ d_c^2 - r^2 s^2 &= k^2 - 2k d_c + d_c^2. \end{aligned} \quad (17)$$

The d_c^2 terms cancel; substituting $d_c = g - Ls$ and collecting terms in s yields a quadratic:

$$r^2 s^2 + 2kLs + (k^2 - 2kg) = 0, \quad (18)$$

with solution

$$s = \frac{-kL \pm \sqrt{k^2(L^2 - r^2) + 2kgr^2}}{r^2}, \quad k = \frac{2\pi\epsilon_0 r^2}{C}. \quad (19)$$

The two roots correspond to positive and negative tilt directions. Only the root satisfying the physical limit (3) is physically meaningful, so

$$\boxed{\alpha_{\text{exact}} = \arcsin\left(\frac{-kL \pm \sqrt{k^2(L^2 - r^2) + 2kgr^2}}{r^2}\right)}, \quad (20)$$

where the sign is chosen such that $|\alpha| < \arcsin(g/(L+r))$.

7 Numerical Example and Graphical Comparison

7.1 Instrument parameters

Parameter	Value	Description
L	40 mm	Pendulum length
r	10 mm	Plate radius
g	0.1 mm	Neutral air gap

From equation (11) the capacitance at $\alpha = 0$ is

$$C_0 = \frac{\pi \times (10^{-2})^2 \times 8.854 \times 10^{-12}}{10^{-4}} = 27.82 \text{ pF}. \quad (21)$$

The physical validity limit from equation (3) is

$$\alpha_{\text{max}} = \arcsin\left(\frac{0.1}{50}\right) = \arcsin(0.002) \approx 412.5'' \approx 2000 \mu\text{m/m}. \quad (22)$$

7.2 Comparison of the three models

Figure 1 shows C as a function of α for the exact formula (9), the tilt-neglecting approximation (12), and the linear approximation (14) over 70 % of the valid range. A secondary axis expresses the angle in $\mu\text{m/m}$ (slope units used in metrology), related to arc seconds by $1'' \approx 4.848 \mu\text{m/m}$.

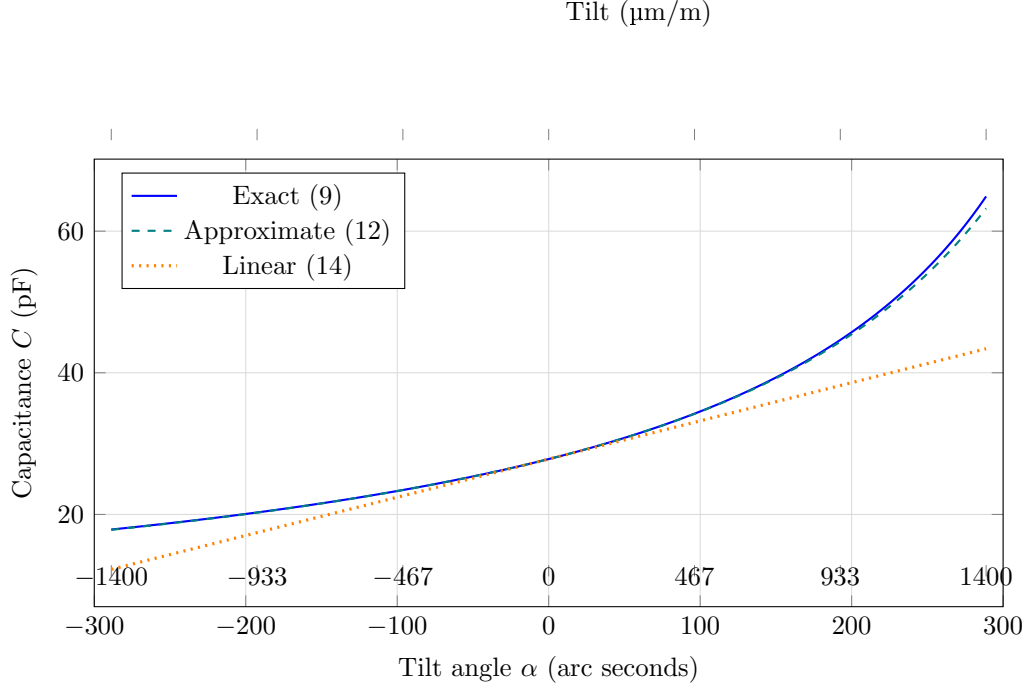


Figure 1: Capacitance versus tilt angle for the three models. Instrument parameters: $L = 40$ mm, $r = 10$ mm, $g = 0.1$ mm. The plot covers 70 % of the physical validity range. The exact and approximate curves are nearly indistinguishable at this scale, confirming that the tilt correction is negligible for these dimensions. The linear model diverges noticeably toward the edges of the range.

8 Summary

Starting from the parallel-plate capacitance formula, three expressions for the capacitance of a pendulum-coupled circular capacitor have been derived and compared.

1. **Exact formula** (eq. 9): valid for all angles within the physical limit (3); accounts fully for both the change in centre gap and the angular tilt of the plates.
2. **Approximate formula** (eq. 12): treats the plates as parallel using only the centre gap; accurate when $r \sin \alpha \ll d_c$. For the example parameters the error is negligible across the entire operating range.
3. **Linear approximation** (eq. 14): a first-order Taylor expansion around $\alpha = 0$; valid only for small deviations from the neutral position.

Each formula has been inverted analytically to yield α as a function of C (equations 20, 16, 15). The exact inversion requires solving a quadratic in $\sin \alpha$; both roots are real and correspond to tilt in opposite directions, and the physically valid root is selected by the limit (3).